

SOFTWARE CONFIGURATION MANAGEMENT PLAN (SCMP)

Project: Planify | **Standard:** IEEE 828-2012 | **Version:** 1.1

1. Purpose & Scope

This Software Configuration Management Plan (SCMP) documents the configuration management activities and controls for Project Planify. Its purpose is to ensure the integrity, traceability, and reproducibility of all software artifacts through disciplined identification, change control, status accounting, and audit practices. The scope of this plan includes the Next.js frontend, the Firebase backend, and all associated project documentation and deliverables.

2. Roles & Responsibilities

- **Configuration Manager:** Oversees the Master Library, establishes and enforces baselines, and coordinates configuration audits.
- **Developers:** Create and manage feature branches, apply unit-level versioning practices, and ensure commits are traceable to change requests or work items.
- **QA Lead:** Plans and executes Functional and Physical Configuration Audits (FCA/PCA) and verifies that changes meet acceptance criteria.
- **Change Control Board (CCB):** Reviews and approves or rejects Change Requests (CRs) and provides governance for release baselines.

3. Configuration Identification

All configuration items (CIs) shall be uniquely identified using the following naming convention:

PLN-[CAT]-[Name]-v[X.X]

Where:

- **PLN:** Project prefix.
- **[CAT]:** Category, e.g., **SRC** (Source), **DOC** (Document), **DB** (Database).
- **[Name]:** Descriptive name of the item.
- **v[X.X]:** Version number following semantic intent for the project.

Examples:

- PLN-SRC-authService-v1.0
- PLN-DOC-SCMP-v1.1
- PLN-DB-schema-v2.0

4. Change Control Process

No direct changes shall be applied to the production **main** branch without an approved Change Request (CR). The formal change control workflow is as follows:

1. **Submission:** Initiator files a CR with justification, affected CIs, and proposed changes.
2. **Impact Analysis:** Development and QA estimate technical, schedule, and risk impacts.
3. **CCB Approval:** The CCB reviews analysis and either approves, rejects, or requests modification of the CR.
4. **Verification in QA:** Upon approval, changes are implemented in development branches and verified in QA/UAT environments.
5. **Merge:** After successful verification and audit, CR changes are merged into **main** as part of a controlled baseline update.

All CRs, decisions, and status updates shall be recorded and maintained in the project's configuration status accounting records.

5. Branching Model

- **main:** Production releases and formal baselines. Only updated through approved merges following CCB approval.
- **UAT/QA:** Integration and acceptance testing branches; used for auditing and user acceptance activities.
- **development:** Continuous integration and integration testing for combined feature work.
- **feat/*:** Topic branches for individual feature development; short-lived and merged into **development** when complete.

Branch protection rules and required review checks shall be enforced on **main** and other critical branches.

6. Baseline Management

Baselines provide controlled snapshots of project assets at defined points in the lifecycle.

- **Baseline 1 (Functional):** Freeze of requirements, design decisions, and the SCMP itself. Used as the reference for implementation activities.
- **Baseline 2 (Product):** Final audited code, configuration items, and fully implemented and approved CRs ready for production release.

Baseline promotion requires verification activities, completion of associated CRs, and sign-off from the Configuration Manager and CCB.

Document Control

- **Prepared by:** Configuration Manager
- **Approved by:** Change Control Board
- **Date:** Version 1.1